

Notation

x is an element of A

$$x \in A$$

x is not an element of A

$$x \notin A$$

Specifying a Set

- Roster Notation
- Set Builder Notation
- Set operations applied to existing sets

The above have the following in common:

- Set description are surrounded by curly brackets

Roster Notation

List elements, separated by commas “,” between ‘{’ and a ‘}’

Example

$$A = \{1, 4, 7, 9\}$$

When communication requires, we can use patterns.

Example

$$B = \{4, 5, 6, 7, \dots, 28\}$$

then $22 \in B$ but $34 \notin B$

Set Builder Notation

{subject | predicate}

- Subject: possibly restricted to a domain

Example

$$C = \{x \mid x \text{ is a student in this room}\}$$

$$D = \{x \in B \mid x \text{ is a multiple of } 6\}$$

Here our subject is a domain restriction,

In this case $D = \{6, 12, 18, 24\}$

Definition

Two sets, S and T are equal, iff they have the same elements.

i.e

$$x \in S \leftrightarrow x \in T$$

In this case we write $S = T$

Example

if $B = \{4, 5, 6, 7, \dots, 28\}$

Let $E = \{x \in B \mid x \text{ is a perfect square}\}$

Let $F = \{4, 9, 16, 25\}$

Let $G = \{16, 9, 16, 4, 16, 16, 4, 25, 16\}$

In this case:

$$E = F$$

$$F = G$$

$$E = G$$

Consequently

$$S \neq T$$

means

$$\exists x((x \in S \wedge x \notin T) \vee (x \in T \wedge x \notin S))$$

Example

$$B \neq E \text{ because } 5 \in B \text{ but } 5 \notin E$$

Common Sets of Numbers

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ Natural Numbers (In this class $0 \in \mathbb{N}$)

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$ Integers

We can write the above in Set Builder Notation

$$\{x \mid x \in \mathbb{N} \vee -x \in \mathbb{N}\}$$

$$\mathbb{Z}^+ = \{x \in \mathbb{Z} \mid x > 0\}$$

$\mathbb{Q}$ = The rational numbers

“Q” is for quotient

$$\left\{x \mid \exists p \in \mathbb{Z}, \exists q \in \mathbb{Z}^+ \left(x = \frac{p}{q}\right)\right\}$$

$$\mathbb{Q}^+ = \{x \in \mathbb{Q} \mid x > 0\}$$

$\mathbb{R}$ = The real numbers

$\mathbb{C}$ = Complex Numbers

$$\{a + bi \mid a \in \mathbb{R} \wedge b \in \mathbb{R}\}$$

Intervals

Continuous subsets of $\mathbb{R}$

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\} \quad [a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

We can mix and match the brackets and square brackets.

$$(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$$

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$$

Example

$$3 \in (2.5, 4.6) \text{ (3 is between 2.5 and 4.6)}$$

$$2.5 \notin (2.5, 4.6)$$

but

$$2.5 \in [2.5, 4.6)$$

A Special Set

The empty set is the unique set with no elements

We write $\emptyset$ instead of $\{\}$ when appropriate

Definition

A set A is a subset of B if every element of A is also an element of B

We write $A \subseteq B$

Alternatively $B \supset A$ read “B is a super of A”

We can write $A \subset B$ A is a strict subset of B

to say $A \subseteq B$ and $A \neq B$

think $\leq$ vs $<$

Note

$A \subseteq B$

means

$$x \in A \rightarrow x \in B$$

implicitly there is a universal quantifier and we interpret this as

$$\forall x(x \in A \rightarrow x \in B)$$

Thus “A is not a subset of B”

means

$$\neg(\forall x(x \in A \rightarrow x \in B))$$

$$\equiv \exists x(x \in A \wedge x \notin B)$$

By De Morgans Law and CD-Equivalence

Recall

$$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$$

$$\equiv \neg\neg p \wedge \neg q$$

$$\equiv p \wedge \neg q$$

Example

If $S = \{\text{red}, \text{blue}\}$

$T = \{\text{red}, \text{blue}, \text{blue}\}$

$R = \{x \mid x \text{ is a colour}\}$

Then $S \subseteq T$ and $T \subseteq R$

but $T \not\subseteq S$ since $\text{green} \in T$ but $\text{green} \notin S$

and $R \not\subseteq T$ since ... (exercise)

Theorem

If S is any set then,

1. $\emptyset \subseteq S$

2. $S \subseteq S$

Proof of (1)

We need to show

$$\forall x(x \in \emptyset \rightarrow x \in S)$$

By contradiction the alternative would be $\exists x(x \in \emptyset \wedge x \notin S)$

But $x \in \emptyset$ is never true

This structure pattern is called a vacuous proof, the implication is true because its hypothesis is never true.

Proof of (2)

... Exercise

Corollary

If S is any set other than the emptyset, then S has at least two subsets.

Definition

If we can count all of the elements of S , and the total is n we say S is a finite set and the cardinality of S is n , we write:

$$|S| = n$$

Example

if $A = \{\emptyset, 1, \{1, 2, 4\}, \{\}, \{1, 1, 2, 4, 2\}, \emptyset, 1\}$

Then $|A| = 3$

Elements are: 1, $\emptyset$, and $\{1, 2, 4\}$

Power Sets

The power set of a set S is the set of all subsets of S

i.e $A \in P(S)$ iff $A \subseteq S$

Example

With A as before:

$$P(A) = \{\emptyset, \{\emptyset\}, \{1\}, \{1, 2, 4\}, \{\emptyset, 1\}, \{\emptyset, \{1, 2, 4\}\}, \{1, \{1, 2, 4\}\}, \{\emptyset, 1, \{1, 2, 4\}\}\}$$

Note:

$$|P(A)| = 8 = 2^3$$

More generally,

$$|P(S)| = 2^{|S|} \text{ if } S \text{ is finite}$$

Cartesian Products

Named after Des Cartes

The set $A_1 \times A_2 \times A_3 \times \dots \times A_n$

consists of all ordered n -tuples:

$(x_1, x_2, \dots, x_n)$ such that $x_i \in A_i$ for each i